Smart Vehicle Safety & Monitoring System

An integrated IoT prototype combining sensors, alerts, and real-time cloud connectivity — built for safer goods transport on every road.

ENGINEERING PROJECT PRESENTATION

U

System Architecture at a Glance

Eight modules work together – from physical load sensing to cloud dashboards – forming a comprehensive vehicle safety ecosystem.

Load & Overload

Load cell + HX711 for real-time weight monitoring

Obstacle Detection

Ultrasonic sensors for animal and load level alerts

Tamper Detection

RFID module linked to Blynk IoT dashboard

Driver Sobriety

MQ3 sensor for alcohol detection + buzzer

Turn Assistance

ESP32-CAM for valley and curved road guidance

Smart Parking

IR sensor at rear for proximity-based parking assist

Load Monitoring & Overload Alert

Hardware Used

- **5 kg Load Cell** with HX711 amplifier module
- **Ultrasonic Sensor** to detect if cargo height exceeds safe level
- **OLED Display** shows live weight and overload warning
- **Buzzer** triggers an audible alert on threshold breach

Weight Readout

HX711 amplifies load cell signal; ESP32 calculates real-time weight in kg and renders it on OLED.

Level Breach

Ultrasonic sensor measures cargo height. If distance falls below threshold, OLED flashes "LEVEL EXCEEDED" and buzzer activates.

Under-Vehicle Animal Detection & Smart Parking

🐾 Animal Detection

A second ultrasonic sensor mounted beneath the chassis detects animals or low obstacles. Triggers buzzer + OLED warning before the driver moves.

🅿 Smart Parking

An IR sensor at the rear detects proximity to walls or vehicles. Provides a proximity alert – ideal for reversing in tight spaces.

Demo Tip: Place an object under the model vehicle and slowly move it – watch the OLED and buzzer respond in real time.

RFID Tamper Detection + Blynk IoT

The cargo compartment is secured using an RFID module. Any unauthorised access attempt is instantly flagged – locally and on the cloud.

Authorised Tag Scanned

Registered RFID card grants access; event is logged to Blynk app with timestamp.

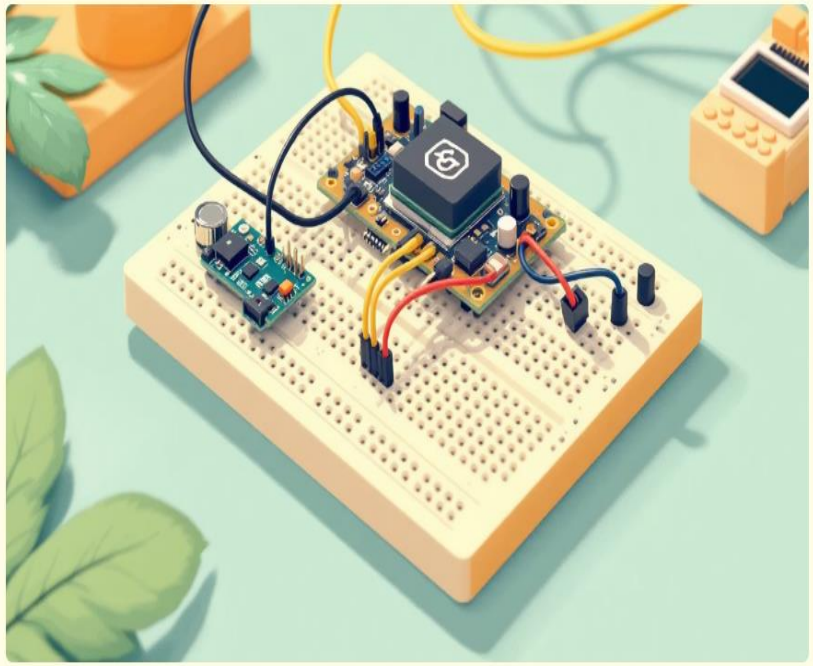
Unknown Tag Detected

Unrecognised card triggers a tamper alert – buzzer sounds and a push notification fires to the Blynk dashboard.

Remote Monitoring

All access events are fetched directly to the Blynk IoT app, enabling fleet managers to monitor cargo security remotely.

Alcohol Detection – MQ3 Sensor



How It Works

The **MQ3 gas sensor** continuously samples the cabin air for alcohol vapour. Readings are processed by the ESP32 and displayed on the OLED in real time.

1

Sense

MQ3 detects ethanol concentration in ambient air

2

Display

Live alcohol value shown on OLED screen

3

Alert

Buzzer triggers if value exceeds safe threshold

Accident Detection – ADXL335 + SIM800L

Design Intent

The **ADXL335 accelerometer** monitors vehicle tilt and sudden G-force changes. On detecting a crash event, the **SIM800L GSM module** is triggered to send an emergency SMS with location data.

- Threshold-based impact detection using X, Y, Z axes
- SIM800L sends SMS alert to registered number
- Potential to integrate GPS coordinates in future

⚠️ Current Status

This module is **under debugging**. Communication between the ADXL335 and SIM800L via ESP32 is not yet stable – likely a power supply or AT command configuration issue.

🔧 Next Steps

Verify SIM800L power requirements (needs steady 2A), check AT command sequence, and test with a dedicated 3.7V LiPo supply.

ESP32-CAM – Turn & Valley Road Assistance

Navigating sharp curves, valley roads, or blind turns is a leading cause of goods vehicle accidents. The ESP32-CAM module provides the driver with a live visual feed to aid safe turning decisions.

Live Video Feed

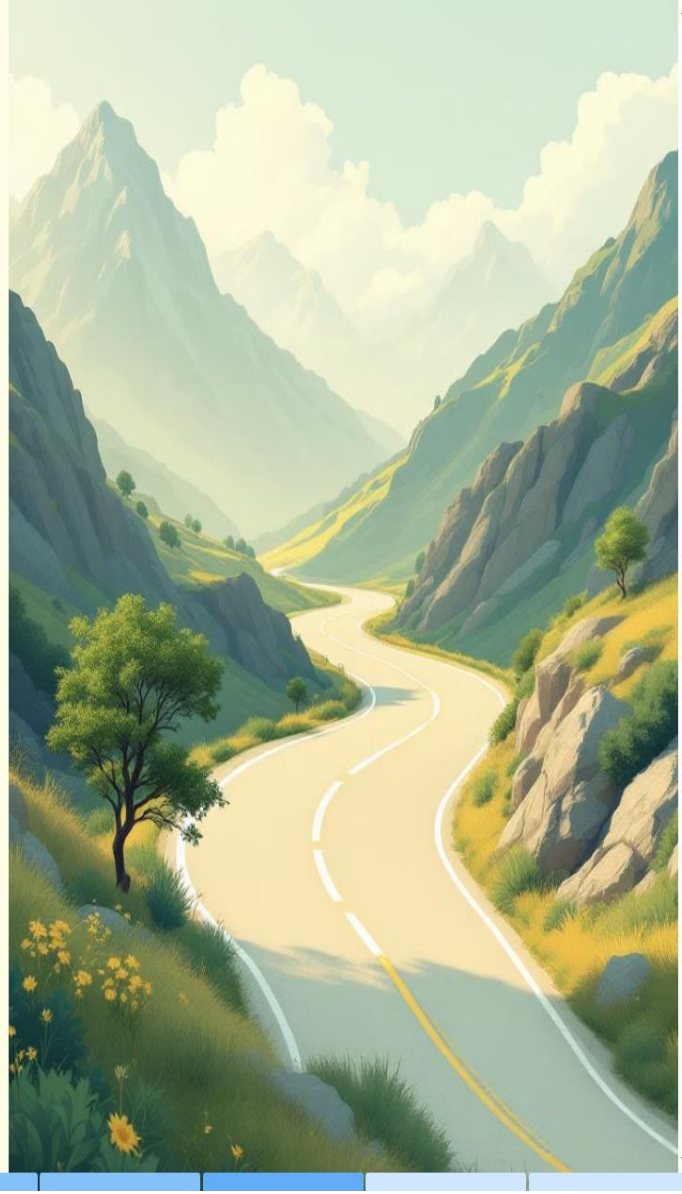
ESP32-CAM streams real-time footage of the road ahead – particularly useful on narrow mountain roads or hairpin bends.

Low-Cost Vision

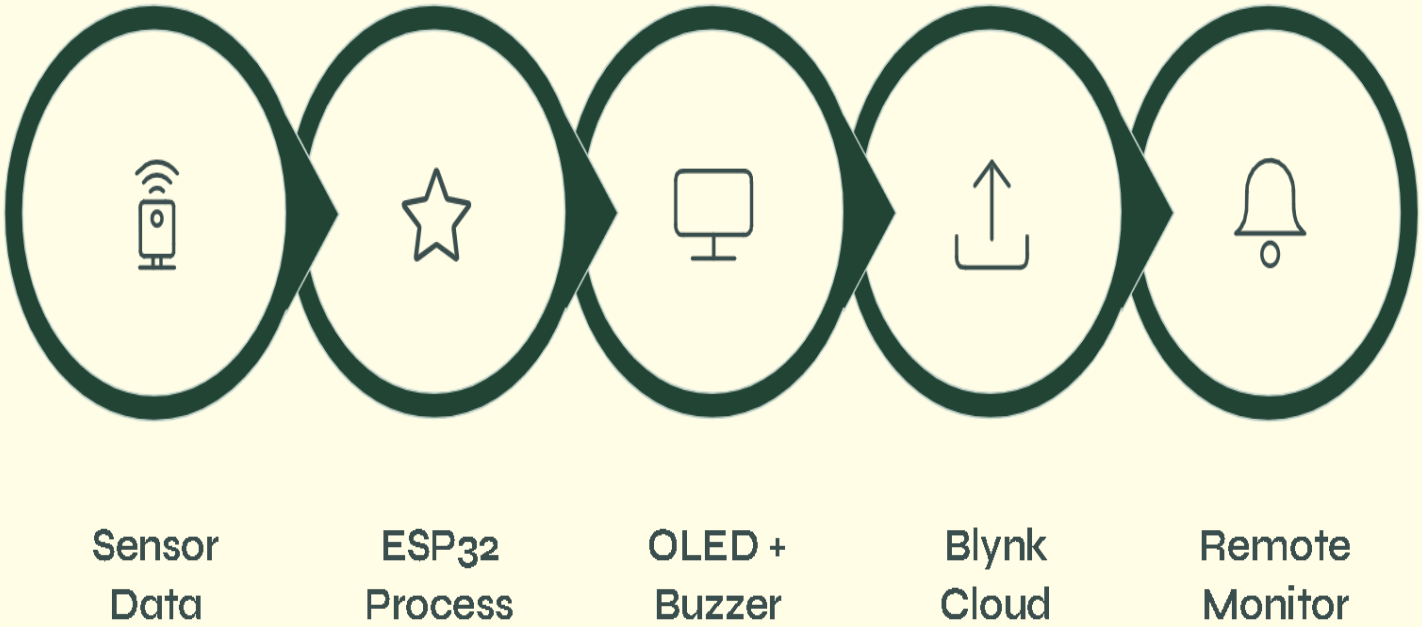
Replaces expensive dashcam setups with an affordable microcontroller-based camera module costing under ₹400.

Future Potential

Can be extended with CV algorithms for lane detection or obstacle recognition using TensorFlow Lite on-device.



Full System Workflow



All sensors feed into the central ESP32 microcontroller. Alerts are issued locally via OLED and buzzer, while critical events – tamper, overload, alcohol – are simultaneously pushed to the Blynk IoT cloud for remote fleet monitoring.

Key Takeaways & Next Steps

Multi-Module Integration

Successfully integrated 7 independent sensor modules into a single ESP32-based vehicle safety platform.

Real-Time Cloud Connectivity

RFID tamper and overload events push live data to the Blynk IoT app – demonstrating practical fleet telematics at low cost.

Pending: Accident Detection

ADXL335 + SIM800L module requires power supply debugging and AT command validation before full deployment.

Scalability

Architecture is modular – each safety function can be independently upgraded, making this a strong foundation for production-grade vehicle IoT systems.